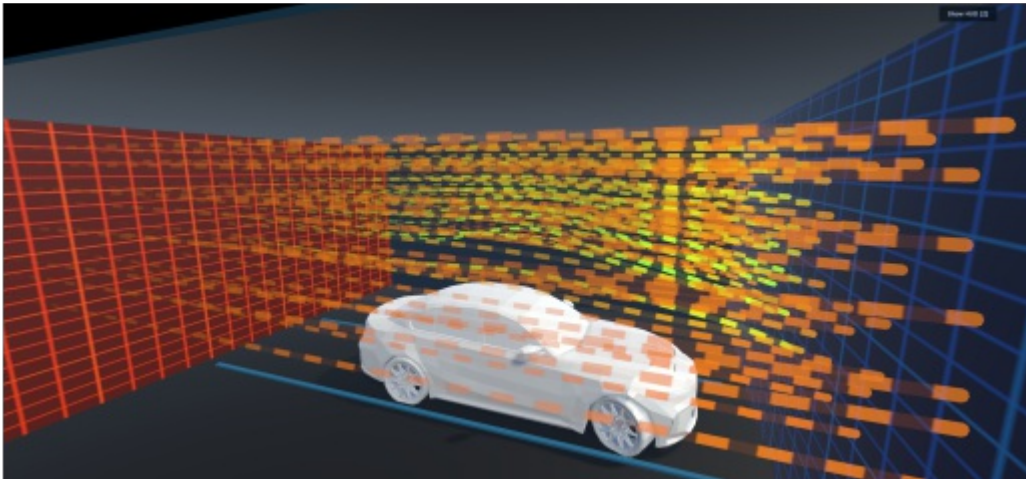




FLUIDWORKS EXTERNAL AERODYNAMICS REPORT

FluidWorks CFD Simulation Report



Model Name: bmw_m3__low-poly_stylized.glb

Date: 2026-03-25 18:28

Software Version: FluidWorks v1.1.0

Accuracy: REGULAR

Generated using FluidWorks CFD Simulation Platform | 2026-03-25 18:28

Page 1



TABLE OF CONTENT

INTRODUCTION 3

GEOMETRY DETAILS 3

SIMULATION SETUP 4

SOLVER DIAGNOSTICS 5

VISUALIZATION RESULTS 6

CONCLUSION 10



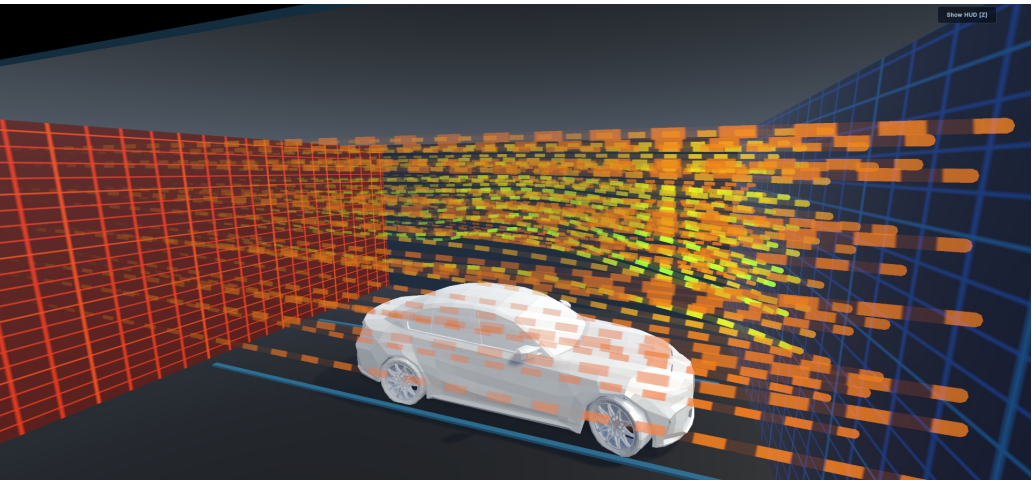
INTRODUCTION

Thank you for choosing FluidWorks to analyze & optimize the aerodynamics of your design. We hope that this report will provide you with the necessary insight. Should you require further assistance, feel free to get in touch with our engineering team.

GEOMETRY DETAILS

These are the dimensions of the bounding box around the object after rotation and scaling.

Parameter	Value
Model Name	bmw_m3__low-poly_stylized.glb
Bounding Box X [m]	LocalZ
Camera Position	(-3.41, 0.83, -1.93)



Primary view of the analyzed geometry



SIMULATION SETUP

Physics Settings	
Viscosity [Pa·s]	0.000018
Density [kg/m³]	1.23
Fluid Type	Air (Newtonian)
Numerical Settings	
Timestep [s]	1.0000
Iteration Count	4
Solver Type	RANS - Steady State



SOLVER DIAGNOSTICS

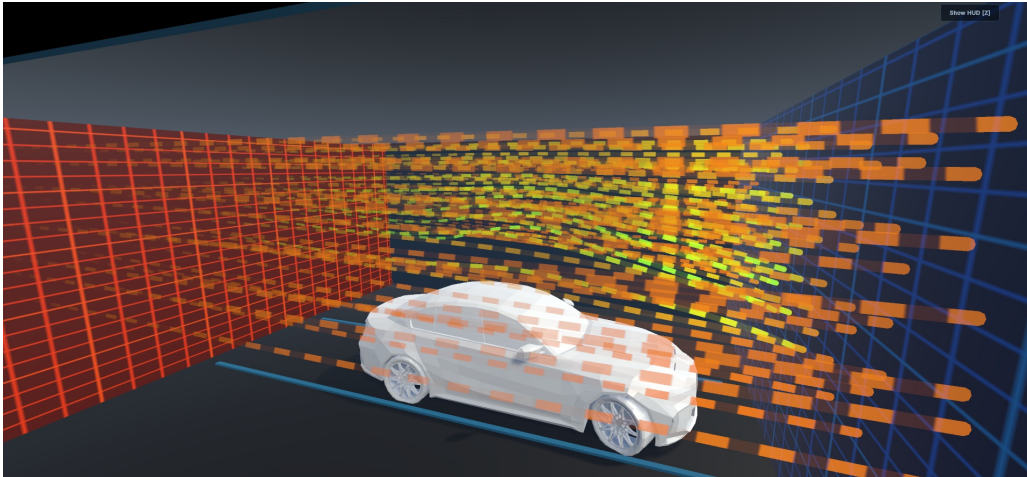
The following metrics provide a quantitative summary of the simulation convergence and performance.

Metric	Result
Mean Velocity [m/s]	43.58
Max Velocity [m/s]	68.38
Pressure Drop [Pa]	0.04
Wall Shear Stress [Pa]	0.0089
Vorticity [1/s]	14.7221
Divergence Error	6.520437



VISUALIZATION RESULTS

FLOW STREAMLINES



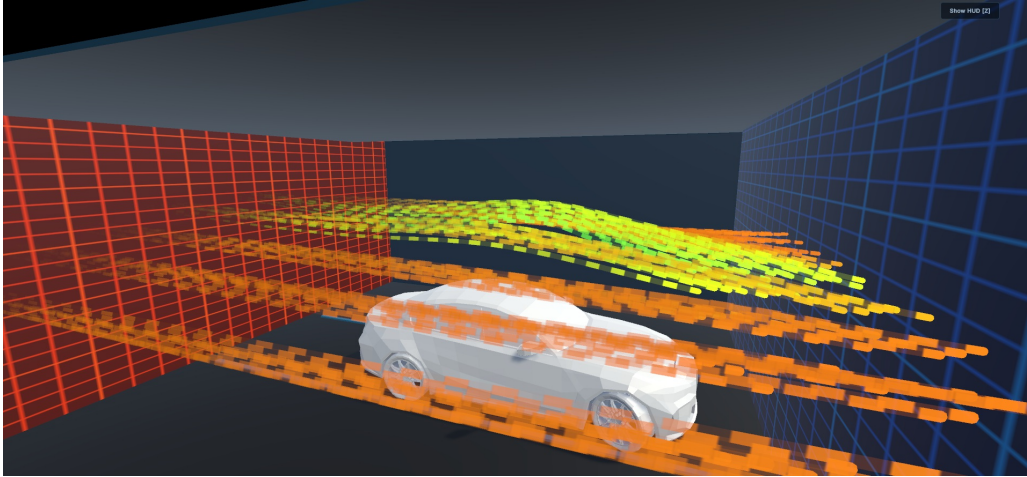
Primary streamline visualization of the active flow field.

Streamlines show the dominant path followed by the flow around the model and help reveal acceleration zones, wake shape, and large-scale separation behavior.

FluidWorks CFD Simulation Report



HORIZONTAL STREAMLINES



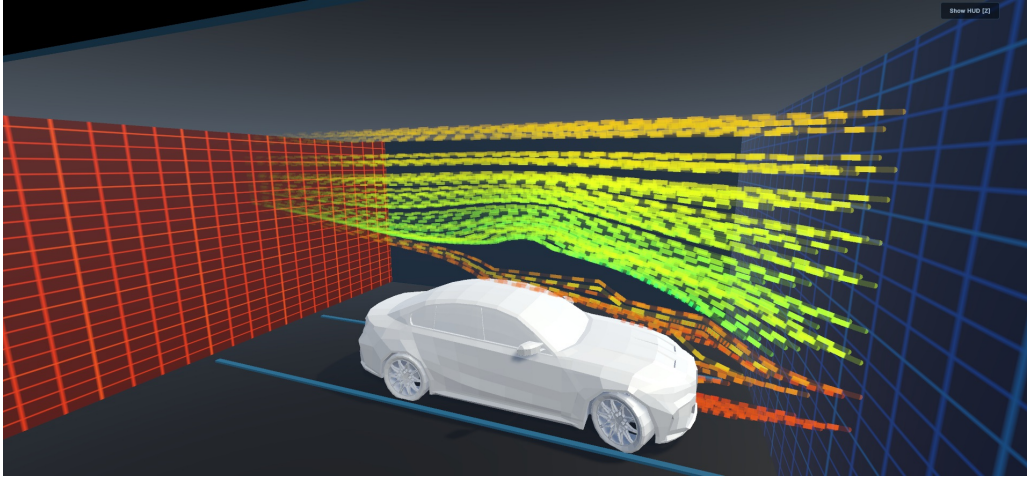
Horizontal streamline pass highlighting lateral flow behavior around the body.

This view emphasizes sidewash, lateral deflection, and horizontal wake development, making it easier to inspect symmetry and side-surface interaction.

FluidWorks CFD Simulation Report



VERTICAL STREAMLINES



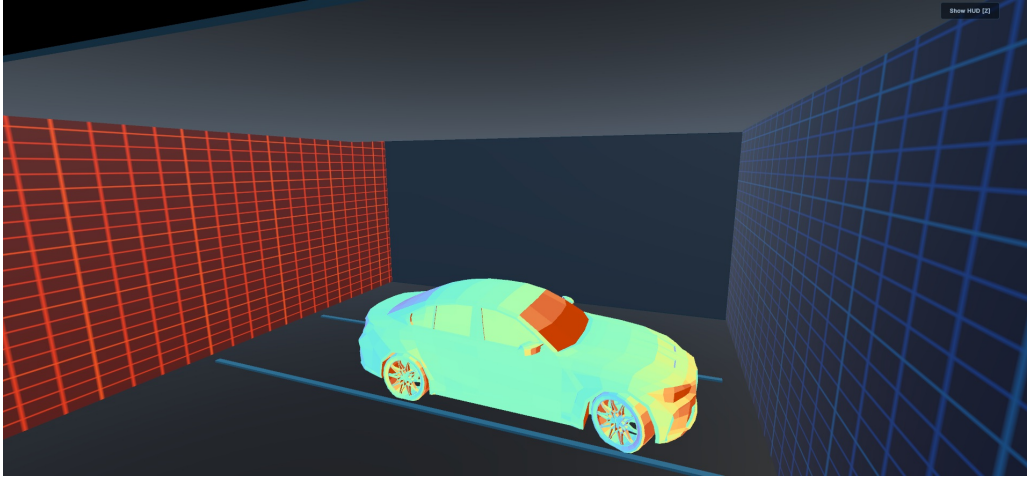
Vertical streamline pass highlighting roof flow, ground interaction, and wake rise.

This view is useful for identifying vertical deflection, roof acceleration, underbody interaction, and the way the wake lifts or settles behind the geometry.

FluidWorks CFD Simulation Report



SURFACE PRESSURE



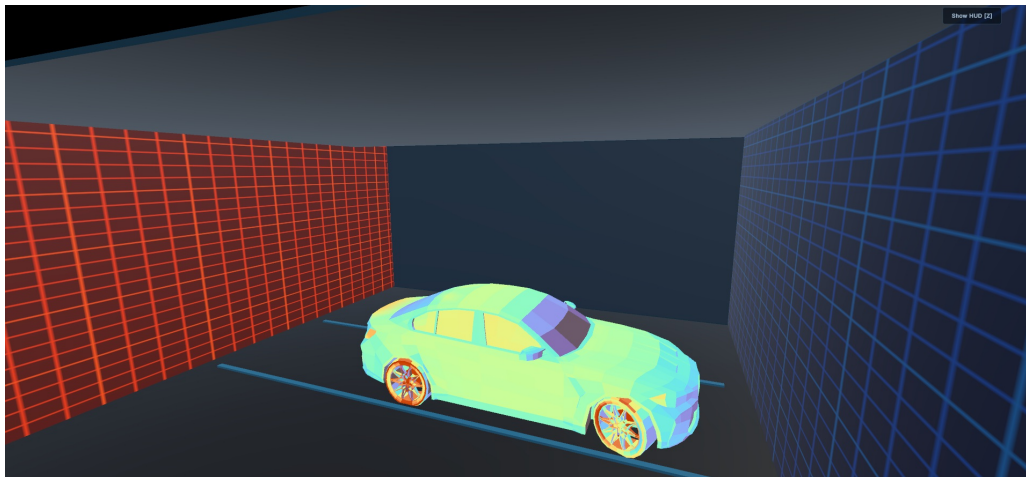
Surface pressure contour mapped directly onto the model body.

Surface pressure highlights stagnation zones, suction regions, and local loading trends that contribute directly to the pressure component of aerodynamic drag and lift.

FluidWorks CFD Simulation Report



SURFACE FRICTION



Surface friction contour showing the wall-shear response on the body.

Surface friction indicates where the flow is strongly attached, where shear intensifies, and where the body is likely experiencing stronger skin-friction effects.



CONCLUSION

Wind Tunnel Assessment: *Solver divergence detected; results may require further refinement. Low pressure drop indicates a highly optimized flow path.*

This report was automatically generated by the FluidWorks CFD Simulation Platform. The results provided are based on the simulation parameters defined in the setup section. For detailed engineering optimization based on these results, please consult with an aerodynamicist.

